

Connecting Numerical, Graphical and Algebraic Ideas of a Limit

Answer Key

Precalculus

Quick Answers

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|---|---|
| 1. 6 | at 0.5 = 0.9589, (c) the limit = 1 |
| 2. (a) $f(3.9) = 3.9748$, (b) $f(4.1) = 4.0248$, (c) the limit = 4 | 7. (a) the limit = $\frac{1}{4}$, — |
| 3. (a) the left-hand limit = 3, (b) the right-hand limit = 4, — | 8. $\frac{1}{6}$ |
| 4. 1 | 9. (a) the sum = 1, — |
| 5. (a) the fourth partial sum = $\frac{45}{8}$, (b) the infinite sum = 6 | 10. (a) the value at $n = 100 = 2.7048$, (b) the value at $n = 10000 = 2.7181$, — |
| 6. (a) the value at 0.1 = 0.9983, (b) the value | 11. (a) the limit = 3, — |
| | 12. (a) the limit = 2, — |
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What is verified

19 of 25 answers machine-checked against SymPy, across 12 problems.
— marks an answer only you can judge — problems 3, 7, 9, 10, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus

HSF-BF.A.2 – problems 5, 9

LIM-1 – problems 1, 2, 3, 4, 6, 7, 8, 10, 11, 12

Difficulty 1–5 of 5 across 25 tagged checks.

Common wrong answers

1. If they got 0: substituted $x = 3$ into the numerator and denominator separately and reported the numerator
 8. If they got 1: cancelled the 9 and the 3 under the radical and read the limit as 1
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1. Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.

Solution: Substitution gives 0/0, so factor: $\frac{(x - 3)(x + 3)}{x - 3} = x + 3$ for $x \neq 3$, and $3 + 3 = 6$.

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2. $f(x) = \frac{x-4}{\sqrt{x}-2}$. (a) $f(3.9)$. (b) $f(4.1)$. (c) the limit at $x = 4$.

Solution (a): $\frac{3.9-4}{\sqrt{3.9}-2} = \frac{-0.1}{-0.0251582}$ 3.9748

Solution (b): $\frac{4.1-4}{\sqrt{4.1}-2} = \frac{0.1}{0.0248457}$ 4.0248

Solution (c): Multiply by the conjugate: $\frac{x-4}{\sqrt{x}-2} \cdot \frac{\sqrt{x}+2}{\sqrt{x}+2} = \frac{(x-4)(\sqrt{x}+2)}{x-4} = \sqrt{x}+2$ for $x \neq 4$, so the limit is $2+2$. 4

3. $g(x) = x+1$ for $x < 2$ and $6-x$ for $x \geq 2$. (a) left-hand limit. (b) right-hand limit. (c) sketch and decide.

Solution (a): Approaching from the left uses the piece $x+1$: $2+1$. 3

Solution (b): Approaching from the right uses $6-x$: $6-2$. 4

Solution (c) — instructor-judged. The sketch shows a line of slope 1 rising to an *open* circle at $(2, 3)$, then a line of slope -1 starting at a *closed* circle at $(2, 4)$ and falling. Since $3 \neq 4$, the one-sided limits disagree and $\lim_{x \rightarrow 2} g(x)$ does not exist. Full credit requires the correct endpoint markings and the "does not exist" conclusion with the two values as its reason.

4. Evaluate $\lim_{x \rightarrow -1} \frac{x^2+3x+2}{x+1}$.

Solution: $x^2+3x+2 = (x+1)(x+2)$, so the quotient is $x+2$ for $x \neq -1$, and $-1+2 = 1$. 1

5. $\sum_{n=1}^{\infty} 3\left(\frac{1}{2}\right)^{n-1}$. (a) S_4 . (b) the whole sum.

Solution (a): $3 + \frac{3}{2} + \frac{3}{4} + \frac{3}{8} = \frac{24+12+6+3}{8}$ $\frac{45}{8}$

Solution (b): $a = 3$, $r = \frac{1}{2}$ and $|r| < 1$, so the partial sums converge to $\frac{3}{1-1/2} = \frac{3}{1/2}$. Note $S_4 = 5.625$ is already close. 6

6. $h(x) = \frac{\sin x}{x}$ in radians. (a) $h(0.1)$. (b) $h(0.5)$. (c) the limit at 0.

Solution (a): $\frac{\sin(0.1)}{0.1} = \frac{0.0998334}{0.1}$ 0.9983

Solution (b): $\frac{\sin(0.5)}{0.5} = \frac{0.4794255}{0.5}$ 0.9589

Solution (c): The values climb toward 1 as x shrinks, and the standard limit confirms it. 1

7. $p(x) = \frac{x-2}{x^2-4}$. (a) the limit at 2. (b) sketch: hole or asymptote?

Solution (a): $\frac{x-2}{(x-2)(x+2)} = \frac{1}{x+2}$ for $x \neq 2$, so the limit is $\frac{1}{4}$. $\frac{1}{4}$

Solution (b) — instructor-judged. The graph is the curve $y = \frac{1}{x+2}$ with an open circle at $(2, \frac{1}{4})$: the factor $x - 2$ cancels, so the function is simply undefined at one point rather than blowing up. It is a *hole*, not a vertical asymptote. Full credit requires the open circle at the right height and the word *hole* (or an equivalent statement about the cancelling factor).

8. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+9}-3}{x}$.

Solution: Multiply by the conjugate:

$$\frac{\sqrt{x+9}-3}{x} \cdot \frac{\sqrt{x+9}+3}{\sqrt{x+9}+3} = \frac{(x+9)-9}{x(\sqrt{x+9}+3)} = \frac{1}{\sqrt{x+9}+3}$$

Substituting $x = 0$ gives $\frac{1}{3+3}$.

$\frac{1}{6}$

9. $\sum_{n=1}^{\infty} (\frac{1}{n} - \frac{1}{n+1})$. (a) the sum. (b) S_N and why the sum is a limit.

Solution (a): The partial sums telescope to $1 - \frac{1}{N+1}$, which tends to 1.

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Solution (b) — instructor-judged. $S_N = (1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + \cdots + (\frac{1}{N} - \frac{1}{N+1})$; every interior term cancels with its neighbour, leaving $S_N = 1 - \frac{1}{N+1}$. The sum of the series is *defined* as $\lim_{N \rightarrow \infty} S_N$, and $\frac{1}{N+1} \rightarrow 0$, so that limit is 1. No finite addition ever reaches 1; the limit is what gives the infinite sum its meaning. Full credit requires the closed form and the statement that the sum is a limit of partial sums.

10. $q(n) = (1 + \frac{1}{n})^n$. (a) $q(100)$. (b) $q(10000)$. (c) name the limit; why is a table not a proof?

Solution (a): $(1.01)^{100}$.

2.7048

Solution (b): $(1.0001)^{10000}$.

2.7181

Solution (c) — instructor-judged. The values are climbing toward $e \approx 2.71828$, and indeed $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$. A table cannot prove it: a table reports finitely many inputs, while a limit is a statement about every input beyond some point. Tables suggest limits; algebra or a theorem establishes them. Full credit requires naming e and giving the finitely-many-values reason.

11. $\lim_{x \rightarrow 2} \frac{x^3-8}{x^2-4}$. (a) the limit. (b) why substitution fails and cancelling is legal.

Solution (a): $\frac{(x-2)(x^2+2x+4)}{(x-2)(x+2)} = \frac{x^2+2x+4}{x+2}$ for $x \neq 2$, and at $x = 2$ that is $\frac{4+4+4}{4} = \frac{12}{4}$.

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Solution (b) — instructor-judged. Substituting gives $\frac{0}{0}$, which is not a value — it only says that $x - 2$ divides both the top and the bottom. Cancelling it produces a new function that agrees with the original at every $x \neq 2$. A limit as $x \rightarrow 2$ never evaluates at 2, only near it, so the two functions have the same limit and the cancellation is legitimate.

Full credit requires both the 0/0 diagnosis and the "agree everywhere except at the point" justification.

12. Synthesis. $r(x) = \frac{x^2-1}{x-1}$ for $x \neq 1$ and $r(1) = 3$. (a) the limit at 1. (b) sketch and explain the discontinuity.

Solution (a): For $x \neq 1$, $\frac{(x-1)(x+1)}{x-1} = x+1$, so the limit is $1+1$. 2

Solution (b) — instructor-judged. The graph is the line $y = x + 1$ with a hole at $(1, 2)$ and a separate solid dot at $(1, 3)$. Continuity at $x = 1$ requires the limit and the function value to be the same number; here the limit is 2 and the value is 3, so r is discontinuous at 1 even though the limit exists. (The discontinuity is removable: redefining $r(1) = 2$ would repair it.) Full credit requires the hole and the dot drawn in the right places and the limit-versus-value comparison stated.